

Report
Of
**Web Development Internship — E-Commerce Platform &
Project Management Dashboard**

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 25SCS1003001305

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Batch: 2025–2029

CANDIDATE'S DECLARATION

I, KULJEET NISCHAL, do hereby declare that the internship report titled "Web Development Internship — E-Commerce Platform & Project Management Dashboard" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: Kuljeet Nischal

Roll No.: 25SCS1003001305

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Web Development Internship Program at CodSoft. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to Mr. Ruman Arshad, Program Manager at CodSoft, and the entire CodSoft team, for sharing their truthful and illuminating views on a number of issues related to this work and for providing me the opportunity to work on the E-Commerce Platform and Project Management Dashboard tasks.

I would also like to thank the faculty of the School of Computer Science and Engineering, IILM University, for their continuous support and guidance throughout this internship.

I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: Kuljeet Nischal

Roll No.: 25SCS1003001305

INTERNSHIP COMPLETION CERTIFICATE

C.ID: 9018130



CERTIFICATE

OF COMPLETION
PROUDLY PRESENTED TO

Kuljeet Nischal

has successfully completed 4 weeks of a virtual internship program in
Web Development

with wonderful remarks at **CODSOFT** from 05/08/2026 to 05/09/2026.

We were truly amazed by his/her showcased skills and invaluable contributions to
the tasks and projects throughout the internship.




Founder



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Date: 08/09/2026

Certificate issued by CodSoft on completion of the 1-month internship period (05 August 2026 to 05 September 2026), C.ID: 9018130.

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4. PROJECT DESCRIPTION

4.1 Introduction

This chapter documents the work completed during a one-month virtual Web Development internship undertaken with CodSoft, an internship and tech-education platform that pairs learners with project-based tasks across domains such as web development, data science, Android development, and Python programming. The internship was completed as part of the academic requirements of the B.Tech CSE programme at IILM University, Greater Noida.

CodSoft's Web Development track offers three levels of tasks — Level 1 (Foundational), Level 2 (Intermediate), and Level 3 (Advanced) — and requires an intern to complete any one level to successfully finish the internship. For this internship, Level 3 was selected, and both of its tasks were completed in full: an E-Commerce Website and a Project Management Tool. Both applications were built using a React.js frontend and a Node.js/Express.js backend, and the complete source code has been published to a public GitHub repository as required by CodSoft's submission guidelines.

This report describes the organization, the problem addressed by each task, the objectives and scope of the work, the technologies used, the system architecture, the development methodology followed, and the outcomes achieved.

4.2 Organization Profile

CodSoft is an internship and tech-education platform focused on turning theoretical programming knowledge into practical, hands-on expertise. It offers virtual internships, project-based learning tracks, and AI-driven career tools such as a resume builder, ATS score checker, AI interview coach, career roadmap generator, and portfolio builder, supporting interns beyond the technical tasks themselves.

The organization is ISO 9001:2015 certified and registered under the Ministry of Micro, Small & Medium Enterprises (MSME), Government of India. Interns are expected to complete assigned tasks with professional diligence, maintain a public GitHub repository of their work, and share a demonstration video of their project on LinkedIn, tagging CodSoft and using the hashtag #codsoft for wider visibility.

The internship offer was issued under Offer ID BY26RY226893, for the position of Web Development Intern, for a duration of one month — from 05 August 2026 to 05 September 2026 — under the supervision of Program Manager Mr. Ruman Arshad.

4.3 Problem Statement

Modern small businesses and independent sellers often need a lightweight, easy-to-understand online storefront through which customers can browse available products, view details, and place orders, without the overhead of a full commercial e-commerce platform. Task 1 addresses this need by implementing a functional, full-stack e-commerce website covering the essential storefront journey — browsing, searching, viewing product details, and checking out.

Separately, small teams frequently coordinate ongoing work using ad-hoc methods such as spreadsheets or chat messages, which do not scale well and make it difficult to track task ownership, deadlines, and progress at a glance. Task 2 addresses this by implementing a project management dashboard that lets a team create projects, break them into assigned tasks, and track each task's status through a simple pipeline.

4.4 Project Objectives

- To gain practical, hands-on experience in full-stack web application development.
- To design and build a REST API using Node.js and Express.js.
- To model and persist application data using MongoDB with Mongoose (Task 1).
- To build responsive, interactive user interfaces using React.js and React Router.
- To implement complete CRUD (Create, Read, Update, Delete) workflows across two independent applications.
- To practise professional software workflows — Git version control, structured repositories, and public code submission.
- To apply theoretical full-stack development concepts learned in coursework to real, working applications.

4.5 Scope of the Project

The scope of this internship was defined by CodSoft's Level 3 Web Development task brief, which required building an E-Commerce Website and a Project Management Tool. The table below summarizes all three available task levels for context; Level 3 was selected to demonstrate full-stack development capability beyond static HTML/CSS/JS tasks.

Level	Difficulty	Tasks
Level 1	Foundational	Portfolio Website, Landing Page, Calculator

Level	Difficulty	Tasks
Level 2	Intermediate	Job Board, Online Quiz Maker
Level 3 (Selected)	Advanced	E-Commerce Website, Project Management Tool

Table 4.1: CodSoft Web Development Task Levels

Within scope: product browsing and search, product detail views, cart and checkout flow, an admin-style product-add form (Task 1); project creation, task assignment, live status updates, and task/project deletion (Task 2). Out of scope for both tasks: real payment gateway integration, user authentication/login systems, and production deployment with a persistent hosted database — these were treated as optional extensions per the task brief and were not required for successful completion of the internship.

4.6 Technologies and Tools Used

Both applications were developed using a common, modern JavaScript-based full-stack toolchain, summarized below.

Category	Technology / Tool
Frontend Library	React.js (with React Router DOM for client-side routing)
Backend Runtime & Framework	Node.js, Express.js
Database (Task 1)	MongoDB with Mongoose ODM
Data Store (Task 2)	In-memory JavaScript data structure
API Style	RESTful APIs (JSON over HTTP)
Styling	Custom, hand-written CSS (no UI framework)
Version Control	Git and GitHub
Development Environment	Visual Studio Code, Node Package Manager (npm)

Table 4.2: Technologies and Tools Used

4.7 System Architecture

Both tasks follow a three-tier architecture consisting of a presentation layer (the React frontend, rendered in the user's browser), an application layer (the Express.js REST API), and a data layer (MongoDB for Task 1; an in-memory store for Task 2). The two architectures are illustrated below.

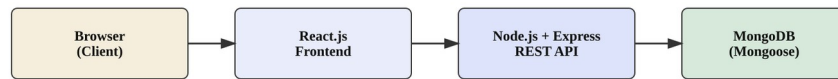


Figure 4.1: Three-Tier System Architecture — Task 1 (E-Commerce Website)

In Task 1, the React frontend requests product data from the Express backend over REST endpoints; the backend queries MongoDB through Mongoose models and returns JSON responses, which the frontend renders as the product catalog, product detail pages, and cart.

Task 1 exposes the following REST API endpoints:

Method	Endpoint	Description
GET	/api/products	Returns all products in the catalog
GET	/api/products/:id	Returns a single product by its MongoDB _id
POST	/api/products	Creates a new product (name, price, category, image, description)

Table 4.3: Task 1 (E-Commerce) — REST API Endpoints

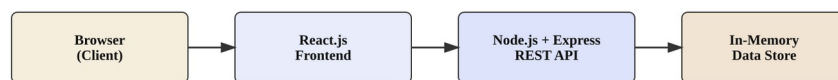


Figure 4.2: Three-Tier System Architecture — Task 2 (Project Management Tool)

In Task 2, the React frontend communicates with the Express backend to create projects and tasks, and to update task status. Project and task data is held in an in-memory array on the server for the duration of the session, keeping the implementation lightweight and easy to reason about.

Task 2 exposes the following REST API endpoints:

Method	Endpoint	Description
GET	/api/projects	Lists all projects
POST	/api/projects	Creates a new project
DELETE	/api/projects/:id	Deletes a project
POST	/api/projects/:id/tasks	Adds a task to a project
DELETE	/api/projects/:pid/tasks/:idx	Deletes a task
PATCH	/api/projects/:pid/tasks/:idx	Updates a task's status

Table 4.4: Task 2 (Project Management) — REST API Endpoints

4.8 Methodology

The development of both applications followed an iterative, feature-driven methodology:

- Requirement analysis: Each task's brief (provided by CodSoft) was studied to identify the required pages, data entities, and user flows.
- Data modelling: For Task 1, a Mongoose schema was designed for the Product entity (name, price, category, image, description). For Task 2, a simple in-memory project/task data structure was designed, with each project holding an array of task objects.
- Backend-first API design: Express routes were built and tested (using tools such as Postman/browser requests) before frontend integration, ensuring each endpoint returned the expected JSON shape.
- Component-based frontend development: The React UI was broken into reusable components/pages — for Task 1: Home, Product Details, Cart, and Add Product; for Task 2: the project dashboard and task cards — each responsible for a single concern.
- Incremental integration: Frontend components were connected to their corresponding API endpoints one at a time, verifying data flow at each step (fetch → render → user action → API call → re-fetch).
- Styling and responsiveness: Custom CSS was applied after functional correctness was achieved, focusing on a clean, readable layout for both desktop and mobile widths.
- Version control: Work was committed to Git incrementally, with the final code organized into a two-project monorepo and pushed to a public GitHub repository.

This approach ensured that at every stage of development a working, testable version of each application existed, reducing the risk of large, hard-to-debug integration issues late in the project.

4.9 Expected Outcomes

The primary expected outcome of this internship was two fully functional, full-stack web applications demonstrating end-to-end command over the React–Node–Express (and, for Task 1, MongoDB) stack. Both outcomes were achieved, as summarized below.

Task 1 — E-Commerce Website ("Aillith"): A working online storefront was delivered with the following features.

Feature	Description
Live Product Catalog	Products fetched from MongoDB via REST; catalog auto-seeds with sample products on first run.
Search & Filter	Real-time client-side search across product name and category.
Product Detail Page	Dynamic route per product showing price, description, and an Add to Cart action.
Shopping Bag & Checkout	Cart state managed in React; running total calculation and a confirmation screen on checkout.
Add Product Form	Admin-style form that POSTs new items to the backend to extend the catalog.

Table 4.5: Task 1 (E-Commerce) — Key Features

Task 2 — Project Management Tool ("Project Hub"): A working project/task dashboard was delivered with the following features.

Feature	Description
Project Initialization	Create a project with title, description, and deadline; each renders as its own card.
Task Breakdown	Each project holds a list of tasks with title, assignee, deadline, and status.
Live Status Updates	Task status (To Do / In Progress / Completed) changes instantly via a dropdown, with colour-coded badges.
Progress Overview	Deadline chips on each project card keep priorities visible at a glance.
Task & Project Deletion	Individual tasks or entire projects can be removed directly from the dashboard.

Table 4.6: Task 2 (Project Management) — Key Features

Beyond the two applications themselves, the internship's expected learning outcomes were also achieved: practical experience with REST API design, MongoDB data modelling, React state management with hooks, multi-page routing, and a complete, professional Git/GitHub submission workflow.

4.10 Certificates and Communication Proof

This section attaches supporting documentation confirming enrolment in the CodSoft internship programme. The Internship Completion Certificate, issued by CodSoft on completion of the internship period, is inserted on page iii of this report. The internship offer letter, confirming the internship position, duration, and Offer ID, is attached in full below.



INTERNSHIP OFFER LETTER

Date : 02/08/2026

ID: BY26RY226893

Dear **Kuljeet Nischal**,

We would like to congratulate you on being selected for the “**Web Development**” virtual internship position with “**CodSoft**”. We at CodSoft are excited that you will join our team. The duration of the internship will be of **1 Month**, starting from **05 August 2026 to 05 September 2026**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects. As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this virtual internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!

Sincerely

Ruman Arshad
Program Manager



ISO Number - TSNUK79053



MSME Regd



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Figure 4.3: CodSoft Internship Offer Letter

5. BIBLIOGRAPHY / REFERENCES

React.js Official Documentation — <https://react.dev>

React Router Documentation — <https://reactrouter.com>

Node.js Official Documentation — <https://nodejs.org/en/docs>

Express.js Official Documentation — <https://expressjs.com>

MongoDB Official Documentation — <https://www.mongodb.com/docs>

Mongoose ODM Documentation — <https://mongoosejs.com/docs/guide.html>

MDN Web Docs (Mozilla) — <https://developer.mozilla.org>

CodSoft Web Development Internship Task Brief, CodSoft (2026) — <https://www.codsoft.in>

Project Source Code Repository — https://github.com/kuljeetnischal/CODSOFT_TASKSNO